Reduced-Pauli entanglement checks for DLA-sector leakage mechanisms on IBM Heron hardware

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(Dated: May 20, 2026)

Abstract

Parity-sector leakage asymmetry in small Kuramoto–XY circuits is a useful hardware observable only if the mechanism can be separated from layout, prepared-state, and readout artefacts. We report a cost-bounded reduced-Pauli entanglement/tomography check for promoted Phase 3 DLA-parity and Fisher-information-modified Kuramoto–XY circuit families. The approved `ibm_marrakesh` execution used physical qubits $[1, 2, 3, 4]$, 162 main circuits, four readout calibration circuits, 2048 shots per main circuit, and 8192 shots per readout circuit. The primary submitted IBM jobs were `d86g7h1789is738vkreg` and `d86ggpis46sc73f6v170`. A preregistered reducer produced 54 observable rows. The mean absolute measured-minus-exact deviation is 0.1299 and the maximum absolute deviation is 0.5561. The largest deviations are concentrated in transverse two-qubit correlators, especially DLA odd-sector `XX`/`YY` edge channels, while the FIM $\lambda_{\mathrm{fim}} = 4$ circuit deviates more strongly than the $\lambda_{\mathrm{fim}} = 0$ reference. A tensor-product readout-mitigation replay leaves the primary mean absolute deviation essentially unchanged (0.1293), and a same-day `ibm_fez` replication gives mean absolute deviation 0.1528, readout-mitigated to 0.1513. A pinned-layout `ibm_fez` repeat with full 16-state readout calibration gives raw mean absolute deviation 0.1494; correlated readout inversion amplifies rather than erases the strongest channels, with the largest amplification still concentrated in transverse edge observables. The result is a mechanism-boundary measurement: it shows structured reduced-Pauli distortions accompanying the small-system DLA/FIM hardware programme, but does not claim scalable tomography, quantum advantage, full-state reconstruction, or backend-general entanglement dynamics.

I. INTRODUCTION

Kuramoto synchronisation and XY spin Hamiltonians provide a compact route from phase-oscillator structure to gate-model quantum dynamics [1, 2]. The dynamical-Lie-algebra language used here follows the standard Lie-algebraic controllability framework for finite-dimensional quantum systems [3]. In the present programme, the relevant small-system hardware question is not whether a broad advantage has been demonstrated. It is narrower: when sector-resolved leakage and retention observables deviate from exact

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Label	Family	Initial state	Depth
`dla_even_shallow`	DLA parity	`0011`	6
`dla_odd_shallow`	DLA parity	`0001`	6
`dla_even_signal`	DLA parity	`0011`	10
`dla_odd_signal`	DLA parity	`0001`	10
`fim_lambda0_reference`	FIM pair	`0011`	4
`fim_lambda4_feedback`	FIM pair	`0011`	4

TABLE I. Promoted source circuits for the reduced-Pauli check.

Kuramoto–XY references on IBM hardware, do reduced-Pauli correlators show an accompanying structure that can constrain the mechanism?

Previous DLA-parity and FIM runs established bounded hardware observations, including backend, layout, state-preparation, and readout sensitivity. A scalar leakage metric alone cannot distinguish coherent correlator deformation from population readout effects or generic depth-dependent decoherence on noisy intermediate-scale quantum hardware [4]. This paper therefore measures a small set of preregistered reduced-Pauli observables for the same class of promoted circuits. Full tomography [5] is unnecessary for this question, and the protocol is closer in spirit to targeted local-observable estimation than to full state reconstruction; related measurement-efficient ideas include classical shadows and local observable prediction [6]. The aim here is narrower still: to compare selected two-qubit correlators against exact classical references under one backend, layout, shot budget, and calibration window.

II. PROTOCOL

The source circuits are listed in Table I. Four DLA-parity circuits compare even and odd initial states at shallow and signal depths. Two FIM-pair circuits compare $\lambda_{\text{fim}} = 0$ and $\lambda_{\text{fim}} = 4$ at fixed initial state and depth.

For each source circuit we measure nine reduced-Pauli basis settings,

$$\{\mathtt{IIXX}, \mathtt{IIYY}, \mathtt{IIZZ}, \mathtt{IXXI}, \mathtt{IYYI}, \mathtt{IZZI}, \mathtt{XXII}, \mathtt{YYII}, \mathtt{ZZII}\}. \tag{1}$$

Each source/basis pair is repeated three times at 2048 shots, giving 162 main circuits. Four computational-basis readout states, `0000`, `0001`, `0011`, and `1111`, are measured at 8192 shots. The complete hardware block therefore contains 166 circuits. The selected physical qubits are $[1, 2, 3, 4]$ on `ibm_marrakesh`; the maximum transpiled depth in the live preflight is 388, and the maximum measurement-basis depth expansion is 1.0718. A second-backend replication on `ibm_fez` used physical qubits $[21, 22, 23, 24]$, the same circuit and shot budget, and jobs `d86h4jlg7okc73el3ra0` and `d86h5d9789is738vlt7g`. A pinned-layout repeat on the same `ibm_fez` qubits added all 16 computational-basis readout calibration states and used jobs `d86hdk8p0eas73dkv9eg` and `d86hedp789is738vm7mg`. Circuit construction, transpilation, and hardware execution used Qiskit and the IBM Quantum platform [7, 8].

III. ANALYSIS

For each measured basis-rotated circuit, the reducer estimates the Pauli expectation

$$\langle P \rangle = \frac{1}{N} \sum_{b} n_b \prod_{i \in \mathrm{supp}(P)} (-1)^{b_i}, \tag{2}$$

where P is the reduced-Pauli label, n_b is the observed count for bitstring b, and N is the total shot count. Repetitions are grouped by source label and basis setting, then compared with exact reference expectations from the committed reference table `data/phase3_entanglement_tomography/entanglement_observable_rows_2026-05-07.csv`.

The primary output rows are committed in `data/phase3_entanglement_tomography/entanglement_tomography_rows_2026-05-20.csv`. The rows have SHA256 digest `b280db120f9f13dfb8dd6b8d4e2e6a86a86feff714e2514be01bd5b3ed95bd83`.

Readout mitigation is applied as a conservative replay, not as a stronger hardware claim. The four calibration circuits are marginalised into independent single-qubit assignment matrices in IBM count-key order, and the tensor-product assignment matrix is inverted before recomputing each reduced-Pauli expectation, following the standard response-matrix/post-processing measurement-error mitigation logic [9]. This is not a full 16-state correlated readout calibration and is not ZNE or PEC. When all 16 calibration states are present in the pinned `ibm_fez` repeat, the reducer instead inverts the full correlated 16-by-16 assignment matrix. This still corrects only readout assignment; it does not correct basis rotations, coherent gate errors, drift, or depth effects.

Circuit label	Mean signed	Mean absolute	Max absolute
`dla_even_shallow`	-0.120474	0.148396	0.314180
`dla_even_signal`	0.061095	0.063458	0.160443
`dla_odd_shallow`	0.098416	0.219708	0.515473
`dla_odd_signal`	0.153497	0.176026	0.556091
`fim_lambda0_reference`	0.030496	0.052205	0.088148
`fim_lambda4_feedback`	0.096107	0.119566	0.274155

TABLE II. Label-level measured-minus-exact deviation aggregates. Each row summarises nine reduced-Pauli observables.

IV. RESULTS

The approved IBM execution completed on `ibm_marrakesh`. The submitted jobs were `d86g7h1789is738vkreg` and `d86ggpis46sc73f6v170`. The analysis produced 54 observable rows with mean absolute deviation 0.1299 and maximum absolute deviation 0.5561 relative to exact references.

Table II gives the label-level aggregates. DLA circuits have larger mean absolute deviation than the FIM pair overall. Within the DLA family, the odd shallow and odd signal circuits produce the largest aggregate departures. Within the FIM pair, $\lambda_{\text{fim}} = 4$ has more than twice the mean absolute deviation of $\lambda_{\text{fim}} = 0$.

Figure 1 shows the signed measured-minus-exact deviations by source label and basis setting. The strongest pattern is not a uniform decay of all correlators. Instead, deviations concentrate in transverse two-qubit channels. The odd-sector DLA signal circuit has large positive shifts in `XXII` and `YYII`; the odd shallow circuit has analogous positive shifts in `IIXX` and `IIYY`. The even shallow circuit moves in the opposite direction on corresponding transverse channels.

The largest individual deviations are shown in Table III. The top four are DLA odd-sector transverse correlators. The strongest FIM-channel departure is the `IXXI` observable for the $\lambda_{\text{fim}} = 4$ circuit.

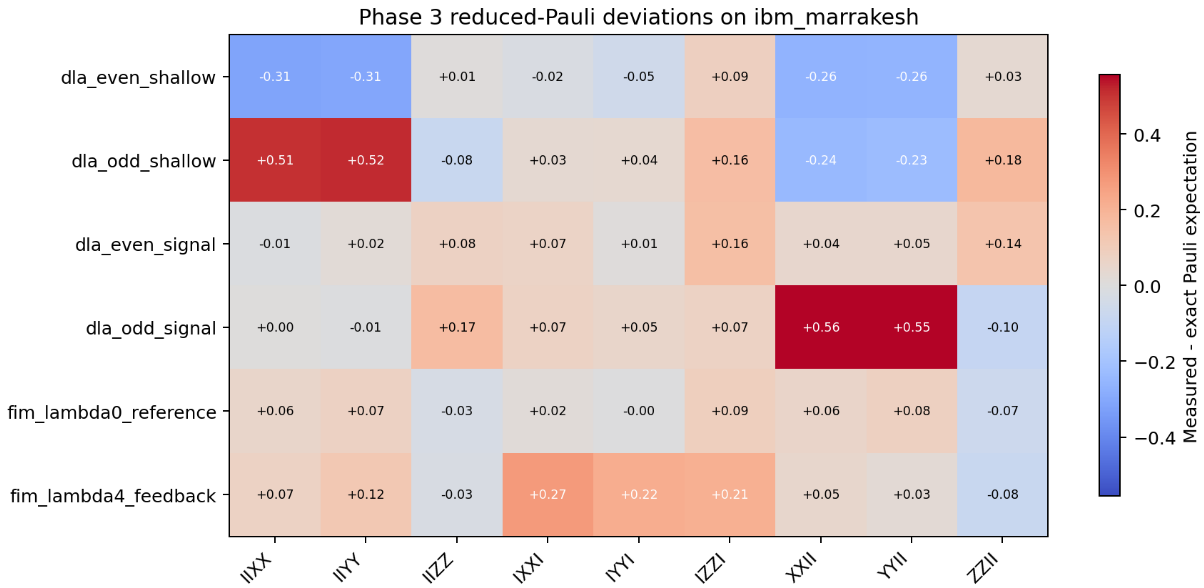

FIG. 1. Measured-minus-exact reduced-Pauli deviations for the completed `ibm_marrakesh` run. The strongest deviations concentrate in transverse edge correlators rather than in all observables uniformly.

Circuit label	Basis	Measured	Exact	Deviation
`dla_odd_signal`	`XXII`	0.431641	-0.124450	0.556091
`dla_odd_signal`	`YYII`	0.429688	-0.124450	0.554138
`dla_odd_shallow`	`IIYY`	0.488932	-0.026541	0.515473
`dla_odd_shallow`	`IIXX`	0.478841	-0.026541	0.505382
`dla_even_shallow`	`IIXX`	-0.456055	-0.141874	-0.314180
`dla_even_shallow`	`IIYY`	-0.447591	-0.141874	-0.305717
`fim_lambda4_feedback`	`IXXI`	0.175781	-0.098373	0.274155

TABLE III. Largest measured-minus-exact deviations in the reduced-Pauli rows.

A. Readout mitigation and same-day replication

Table IV summarises the primary `ibm_marrakesh` run, the same-day `ibm_fez` replication, and the pinned-layout `ibm_fez` repeat with full readout calibration. The tensor-product readout inversion does not collapse the deviations toward zero on either backend. On `ibm_marrakesh`, the mean absolute deviation changes from 0.1299 to 0.1293; on `ibm_fez`, it

Backend	Mean abs.	Max abs.	Mitigated mean abs.	Mitigated max abs.
`ibm_marrakesh`	0.129893	0.556091	0.129304	0.564834
`ibm_fez` auto layout	0.152821	0.479015	0.151317	0.486875
`ibm_fez` pinned full readout	0.149445	0.473481	0.195282	0.991278

TABLE IV. Cross-backend reduced-Pauli deviation summary. Mitigation is independent single-qubit readout inversion for the first two rows and full correlated readout inversion for the pinned full-readout row.

changes from 0.1528 to 0.1513. The pinned-layout repeat gives a similar raw mean absolute deviation, 0.1494, while full correlated readout inversion increases the aggregate mitigated deviation to 0.1953 and the largest mitigated deviation to 0.9913. This does not prove backend-general dynamics, but it does rule out the simplest interpretation that the primary observation is only a single-backend population-readout artefact. The full correlated inversion result is treated as a sensitivity bound, because matrix inversion can amplify calibration and shot noise in individual channels.

The amplification is not uniform across the reduced-Pauli table. In the pinned full-readout replay, transverse edge channels have mean absolute amplification 0.1110 and maximum amplification 0.5178, while transverse-middle channels have mean absolute amplification 0.0026. The two largest amplified rows are `dla_odd_shallow/IIYY` and `dla_odd_shallow/IIXX`, both already in the dominant DLA transverse-edge family. The `ZZ` edge class also shows sensitivity, with mean absolute amplification 0.0946 and maximum amplification 0.2855. Thus the correlated-inversion replay should not be read as a clean mitigation result, but it also does not look like a featureless uniform noise blow-up. The supplementary amplification rows are generated in `data/phase3_entanglement_tomography/entanglement_tomography_full_readout_amplification_top_2026-05-20.csv` and `data/phase3_entanglement_tomography/entanglement_tomography_full_readout_amplification_classes_2026-05-20.csv`.

V. DISCUSSION

The main result is not that the hardware reproduces the exact reduced-Pauli reference structure. It does not. The contribution is that the departures from the exact reference are large, structured, and concentrated in specific correlator channels. This turns the earlier leakage/retention observations into a sharper hardware-mechanism question: the same circuit families that show sector-sensitive survival behaviour also show measurable correlator-level distortions.

The DLA circuits dominate the strongest deviations. The sign structure is particularly important: odd-sector circuits show large positive transverse edge-correlator shifts, whereas the even shallow circuit shows large negative shifts on corresponding channels. This is not consistent with a single scalar decoherence factor multiplying every observable equally. It is more naturally read as a sector- and edge-dependent deformation under the selected backend, layout, circuit family, and calibration window.

The FIM result is more muted but still useful. The $\lambda_{\rm fim} = 4$ feedback circuit deviates more strongly than the $\lambda_{\rm fim} = 0$ reference. This is consistent with the earlier negative fixed-FIM hardware result: the tested digital FIM modification does not simply protect the target structure and introduces a measurable reduced-Pauli distortion channel. The result motivates adaptive FIM follow-up only under a conservative claim boundary; it does not rescue a fixed-$\lambda_{\rm fim}$ protection claim.

The readout boundary remains central. Four readout calibration states are included in each execution block, and independent single-qubit readout inversion leaves the aggregate deviation scale intact. Because the largest deviations occur in transverse basis-rotated measurements, not only in computational-basis `ZZ` channels, the result is unlikely to be a pure population-readout story. It may still include basis-rotation error, layout-dependent coherent error, calibration drift, and depth-dependent decoherence. The safe interpretation is therefore mechanism-boundary evidence, not full causal attribution.

The two `ibm_fez` executions also separate layout robustness from same-layout repeatability. The auto-layout `ibm_fez` run and the pinned-layout repeat use the same physical qubits, $[21, 22, 23, 24]$, and their raw aggregate deviations are close: 0.1528 and 0.1494. The remaining quantitative differences may reflect calibration-window drift, basis-rotation error, coherent layout-specific error, or finite-shot variation. Because the same-order transverse deviations persist across `ibm_marrakesh` and `ibm_fez`, and across two `ibm_fez` executions, a single layout-only explanation is not sufficient; however, layout-specific coherent error remains a plausible contributor to the detailed channel amplitudes.

Prepared label	Dominant observed bitstring	Retention
`0000`	`0000`	0.981079
`0001`	`1000`	0.975342
`0011`	`1100`	0.977661
`1111`	`1111`	0.974365

TABLE V. Readout calibration dominant-state retention for the completed `ibm_marrakesh` execution. The largest reduced-Pauli deviations in Table III are roughly an order of magnitude larger than this dominant-bitstring readout leakage scale.

A. Readout-sensitivity check

The four readout calibration circuits provide a scale check for the largest reduced-Pauli deviations. The dominant measured bitstring for each calibration state is shown in Table V. Because Qiskit reports classical bitstrings with endianness opposite to the source labels for some prepared states, the table reports the dominant observed bitstring rather than treating the printed label as a direct left-to-right comparator. The mean dominant-state retention across the four calibration circuits is 0.9771, corresponding to a mean leakage from the dominant calibration bitstring of 0.0229. The weakest calibration state retains 0.9744 of shots.

This calibration block does not remove the need for caution: transverse basis-rotated measurements can be affected by basis-rotation and coherent gate errors before readout. It does, however, make a pure computational-basis readout-retention explanation implausible for the largest observed effects. The maximum reduced-Pauli deviation, 0.5561, is about 21.7 times the worst dominant-state calibration leakage of 0.0256 and about 24.3 times the mean dominant-state calibration leakage of 0.0229. The paper's claim boundary is therefore refined as follows: readout calibration does not by itself explain the largest transverse correlator deviations, but the result remains a compound hardware-window observation rather than an isolated entanglement mechanism attribution.

VI. CLAIM BOUNDARY

The supported claims are intentionally narrow. The run measured preregistered reduced-Pauli correlators for small Kuramoto–XY DLA and FIM circuit families on one IBM Heron backend and layout. The measured correlators deviate from exact references in a structured way, with the largest deviations in transverse edge channels. This constrains the mechanism interpretation of the earlier DLA/FIM hardware programme.

The result does not support quantum advantage, scalable tomography, full-state reconstruction, or claims about unmeasured circuits and depths. The `ibm_fez` replication supports a limited two-backend robustness check, not backend-general entanglement dynamics.

VII. REPRODUCIBILITY

The source code and data are in the `anulum/scpn-quantum-control` repository. The execution and analysis use Qiskit and IBM Quantum access paths consistent with the cited platform references [7, 8]. The relevant commands are:

- live readiness and gated submission: `scripts/phase3_entanglement_tomography_ibm.py`;
- completed raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_marrakesh_2026-05-20T004334Z.json`;
- completed replication raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_fez_2026-05-20T014536Z.json`;
- completed pinned full-readout raw-count artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_fez_2026-05-20T020452Z.json`;
- post-run reducer: `scripts/analyse_phase3_entanglement_tomography.py`;
- paper-asset generator: `scripts/generate_phase3_entanglement_paper_assets.py`.
- ZNE-subset readiness artefact: `data/phase3_entanglement_tomography/entanglement_tomography_live_ibm_fez_2026-05-20T022653Z.json`.

VIII. CONCLUSION

A preregistered 166-circuit IBM Heron run measured 54 reduced-Pauli observables for promoted DLA and FIM Kuramoto–XY circuits. The resulting deviations from exact references are large and structured, with the strongest departures in transverse DLA edge correlators and a smaller but clear FIM $\lambda_{\mathrm{fim}} = 4$ distortion. The paper's contribution is a bounded mechanism-separation result: it localises correlator-level hardware deformation accompanying the earlier leakage/retention observations, shows that the largest effects exceed the dominant computational-basis readout leakage scale, and verifies that independent readout inversion and a second Heron backend do not remove the aggregate deviation pattern. A pinned-layout repeat with full readout calibration further shows that correlated readout inversion amplifies rather than eliminates the strongest deviations. The next experimental step is therefore not a broader claim, but a preregistered noise-scaling lane on the same reduced-Pauli observables: a small ZNE subset covering the dominant DLA transverse channels and one FIM control channel, with depth-expansion and cost gates fixed before submission. The present paper still avoids scalable tomography, backend-general, full-causal-entanglement, or advantage claims.

The preregistered ZNE follow-up has been preflighted as a 61-circuit subset on `ibm_fez` qubits [21, 22, 23, 24]: five selected observables, three repetitions, global unitary-folding scale factors $1, 3, 5$, and all 16 readout-calibration states. Under an explicit ZNE depth gate of 1900, the maximum transpiled depth is 1684 and the estimated QPU time is 0.56 minutes. This is a readiness result only; no ZNE hardware data are included in the present manuscript. The proposed extrapolation lane follows the established short-depth error-mitigation and digital ZNE literature [10–13].

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